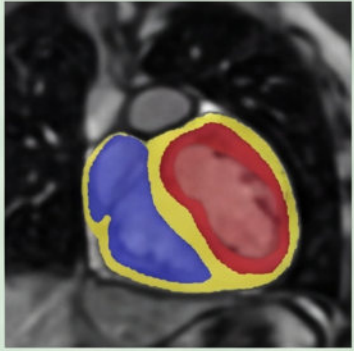


Post-hoc AI Pipeline for auditing cardiac MRI semantic measurements & rules

Formal representation and input space

Native cine CMR
ED/ES images
and masks



Left ventricle,
myocardium

Formal matrix **A**

128 deterministic
image and mask
descriptors

$$\mathbf{A} \in \mathbb{R}^{m \times 128}$$

where m is the
number of patients

Semantic transition and reconstruction phase

Training-only
measured matrix **B**
(28 CMR semantics)

Regularized
transition **T**

Ridge regression

$$\mathbf{T}_\lambda = (\mathbf{A}_z^T \mathbf{A}_z + \lambda \mathbf{I})^{-1} \mathbf{A}_z^T \mathbf{B}_c$$

Reconstructed
semantic
matrix

$$\hat{\mathbf{B}}$$

Examples:
LVEF, RVEDV,
LV myocardial mass,
ED wall thickness

Training-only
measured matrix **B**
(28 CMR semantics)

Conflict-aware rule inference and auditing

Discretization
and symbolic
states **S**

Weighted
entropy-density
discretization
(WEDD)

Rulebook $\mathcal{R}$
(lower approximation granules)

$\mathcal{R}$

Conservative
inference

$$o(\mathbf{x}) = \text{abstain if } \Gamma(\mathbf{x}) = 1$$

Exact match and
clinical alignment
(e.g., dilated
cardiomyopathy proxy)

Contradictory consequents
(Conflict)

Reason-coded abstention
(Missing prerequisites,
unstable reconstruction)